

CS 411 Stage 1: Project Description

Project: Virtually Integrated Agenda

Team 001 - TableForFour

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1 Project Summary

The Virtually Integrated Agenda, or **VIA**, is a centralized event management and logistics platform designed specifically for the Electrical and Computer Engineering (ECE) department at UIUC. Currently, Registered Student Organizations (RSOs) operate individually, relying on scattered communication channels like email lists, Instagram posts, and Discord servers to publicize events. This fragmentation leads to low student engagement and logistical conflicts, such as double-booked rooms or competing major events.

Our platform solves this by providing a unified "Public Feed" where students can discover events across all ECE RSOs in one place. Simultaneously, it provides RSO administrators with a "Logistics Dashboard" that simplifies event creation. This will improve coordination between RSOs and foster better engagement among the ECE student community.

2 Creative Component

Instead of the traditional "blind booking" model where organizers guess which rooms might be available, our project utilizes an **Intelligent Venue Recommender System**. Our system aims to use smart scheduling logic to facilitate booking rooms for the various events by displaying relevant information to the booker.

When an RSO administrator creates an event, they input constraints such as *Expected Attendance*, *Required Room (if any)*, and *Time Window*. The system then performs a complex cross-reference against a database of scraped university course schedules and existing internal reservations. It returns a ranked list of available venues that optimize for capacity utilization (preventing a 10-person meeting from booking a 200-person lecture hall) and conflict avoidance. This requires advanced SQL queries.

3 Usefulness

Basic Functionality

- **Students (Public Users):** Can browse a centralized calendar, filter events by category (such as "Workshop," "Social," "Corporate") and tags (such as "ECEx Event" and "Weekly ECE Socials"), and RSVP to events to build a personal schedule.
- **RSO Administrators:** Can manage their organization's profile, publish events, track attendance metrics, utilize the scheduler tool to secure venues, and attach relevant advertisement for each event for the Kiosk.
- **Kiosk Mode:** A specialized view designed for digital signage in the ECE Building atrium that cycles through upcoming events in real-time.

Comparison to Existing Solutions

While generic calendar applications (like Google Calendar, Apple Calendar) exist, they lack the specific relational context of the university and can't enforce RSO-specific permissions. *IlliniSpaces* provides room availability but does not allow for event management or publicizing. On the other hand, **VIA** combines the utility of a room scheduler with the social discovery aspects of an event platform, tailored specifically for ECE students. As leaders in ECE RSOs, we have seen firsthand how frustrating the current system is, and would use this application in collaboration with other RSOs (namely, iRobotics, Pulse, ECESAC, Solar Car, IEEE, WECE, and HKN). This project aims to solve the collaboration and scheduling problems that we have run into during our time here, and cover the existing weak spots in the systems we use now.

4 Realness: Data Sources

To ensure our ”**Intelligent Scheduler**” operates on realistic constraints, we will integrate three distinct data sources. This combination allows us to model the actual availability of ECE resources.

Dataset 1: UIUC Course Schedule (Real Data)

- **Source:** UIUC Course Explorer API (publicly available XML/JSON endpoints).
- **Content:** A comprehensive list of all Spring 2026 course sections, including *Building*, *Room Number*, *Start Time*, *End Time*, and *Days of Week*.
- **Scale:** This dataset contains over **15,000 course instances**.

Dataset 2: Facilities Reservation Records (Real Data)

- **Source:** Departmental Facility Management Logs (CSV Export).
- **Content:** Historical and current room reservations that are *not* academic courses. This includes exam bookings, one-off seminars, and existing RSO recurring meetings.
- **Utility:** By using this secondary layer of availability data, our application prevents ”false positives” where a room looks free according to the Course Explorer but is actually booked by administration or another organization.

Dataset 3: RSO Event & Activity Log (User-Inputted/Seeded)

- **Source:** Application User Input (Seeded for demonstration).
- **Content:** The transactional data created by our users, including *Event Details*, *RSO Profiles*, and *RSVP Lists*.
- **Implementation:** While the production system relies on live user input, for the purpose of the project demonstration and query testing, we will seed this table with a simulated semester’s worth of events (approximately 100-300 active events) to demonstrate the conflict detection logic against Datasets 1 and 2.

5 Functionality Description

1. **User Registration (INSERT):** Users create accounts linked to their NetID. The system checks against an allowlist (if applicable) and inserts a new row into the `Users` table.
2. **Event Creation (INSERT):** RSO Admins submit a form with event details. The system creates a new `Event` entity and potentially multiple `RoomReservation` entities based on the selection.
3. **RSVP Management (UPDATE/INSERT):** Students toggle their attendance status. This updates the RSVP many-to-many link table.
4. **Event Modification (UPDATE):** RSO Admins can change the time or location of an event. This triggers a constraint check to ensure the new slot is not conflicting.
5. **Event Deletion (DELETE):** RSO Admins can delete the event from the calendar if it gets canceled. This should have a cascading effect for any views, or other data tables that may be using that entry.
6. **Discovery (SEARCH):** Users can search for events using complex filters (such as "Show me all Free Food events on Fridays in ECEB").

6 UI Mockup

Below is a mockup of the **VIA Public Feed**, demonstrating the filtering sidebar and event cards.

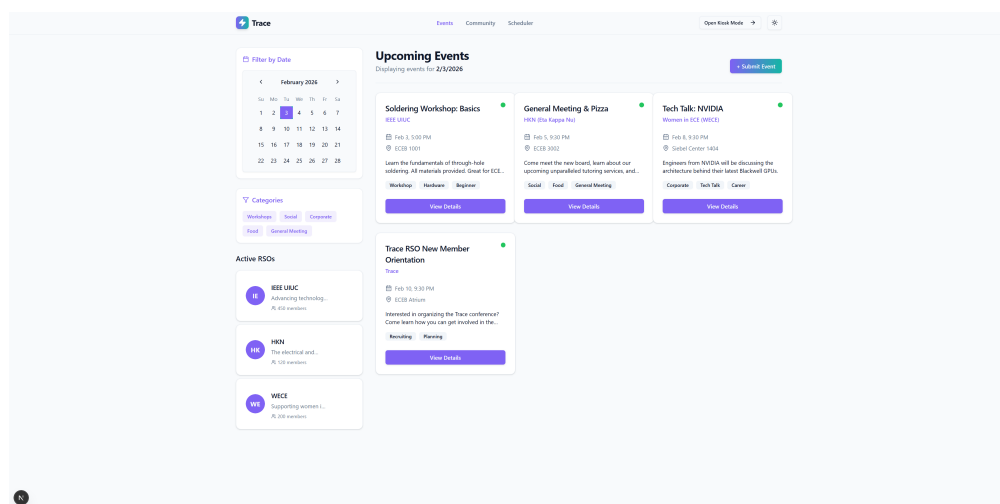


Figure 1: Mockup of VIA front page interface

7 Project Work Distribution

While the entire team will collaborate on the core Database Design, E-R Modeling, and writing the final SQL queries, the specific implementation workload will be distributed as follows:

- **Max Bromberg & Jonathan Martinez Herrada:**

- Primary responsibility for **Backend Architecture**. This includes setting up the database instance (GCP/Local), implementing the Python/FastAPI endpoints, and managing data ingestion from the Course Explorer API.
- Leading the physical database schema implementation

- **Zainab Memon & Meenakshi De:**

- Primary responsibility for **Frontend Development** and UI/UX Design. This involves building the interactive dashboard, the "Public Feed" views, and the Kiosk Mode visualization.
- Designing the specifications for the Advanced Queries to ensure the backend retrieves exactly the data needed for the frontend features such as filtering logic and aggregation requirements.

Note: All four members will contribute to writing the final complex SQL queries and optimizing database performance.